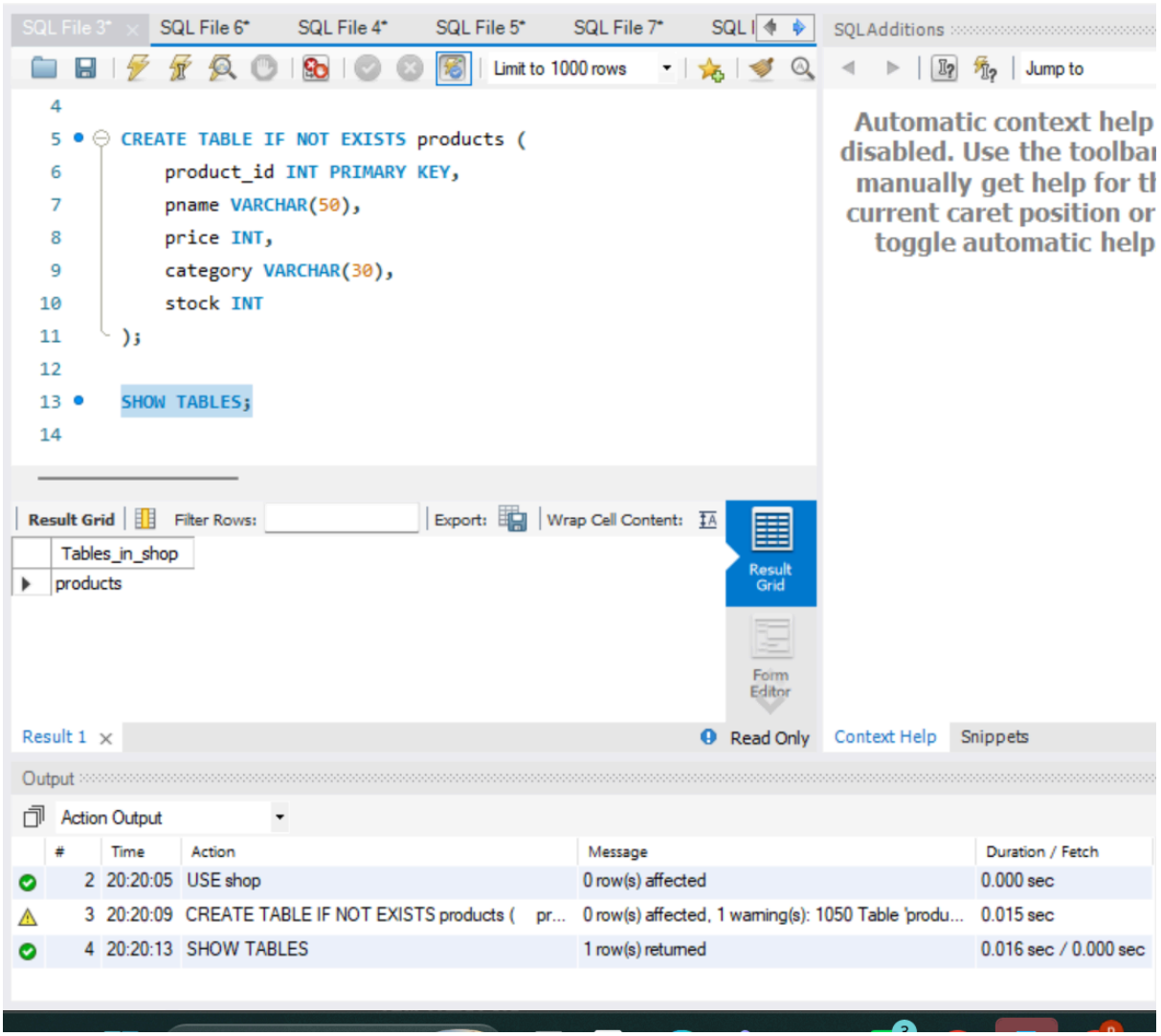
Practice Set 1 – Shop Database

Q1. Create the `shop` database and `products` table. Confirm with `SHOW TABLES`.



Q2. Insert all 6 products in one single `INSERT` statement.

```
INSERT INTO products
VALUES
(1, 'Rice Bag', 2200, 'Grocery', 50),
(2, 'Laptop', 85000, 'Electronics', 10),
(3, 'Mobile', 30000, 'Electronics', 25),
(4, 'Noodles', 25, 'Grocery', 500),
(5, 'Shampoo', 350, 'Beauty', 80),
(6, 'TV', 55000, 'Electronics', 8);
```

5 20:24:26 INSERT INTO products VALUES (1, 'Rice Bag', 2... 6 row(s) affected Records: 6 Duplicates: 0 Warni... 0.032 sec

Q3. Insert product 7 with only product ID and name (`'Pen'`). Which columns become `NULL`?

```
25 •  INSERT INTO products (product_id, pname)
26    VALUES (7, 'Pen');
```

```
6  20:37:47  INSERT INTO products (product_id, pname) VAL...  1 row(s) affected  0.
```

Q4. Try inserting another product with `product_id = 3`. Write the error and explain why.

```
31 •  INSERT INTO products
32    VALUES (4, 'Keyboard', 2000, 'Electronics', 20);
33
34 •  SELECT * FROM products;
35
36
```

Result Grid | Filter Rows: | Edit: | Export/In

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
6	TV	55000	Electronics	8

```
10  20:43:46  SELECT * FROM products LIMIT 0, 1000  7 row(s) returned
```

Q5. Insert two more products in one statement: one Grocery and one Beauty.

```
36
37 •  INSERT INTO products
38    VALUES
39    (8, 'Biscuit', 100, 'Grocery', 100),
40    (9, 'Face Wash', 500, 'Beauty', 60);
41
```

```
11  20:50:45  INSERT INTO products VALUES (8, 'Biscuit', 100...  2 row(s) affected Records: 2  Duplicates:
```

Q6. Show all columns of all products.

```
42 •  SELECT * FROM products;
```

Result Grid | Filter Rows: | Edit:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
6	TV	55000	Electronics	8

products 3

```
12  20:52:47  SELECT * FROM products LIMIT 0, 1000  9 row(s) returned
```

Q7. Show only `pname` and `price`.

```
42 •  SELECT * FROM products;
43
44 •  SELECT pname, price
45    FROM products;
46
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

pname	price
Rice Bag	2200
Laptop	85000
Mobile	30000
Noodles	25
Shampoo	350
TV	55000

products 4

```
13  20:55:39  SELECT pname, price FROM products LIMIT 0, 1...  9 row(s) returned
```

Q8. Show products with prices above 1000.

```
47 •  SELECT *
48    FROM products
49    WHERE price > 1000;
50
```

Result Grid | Filter Rows: | Edit: | Export/

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
6	TV	55000	Electronics	8
NULL	NULL	NULL	NULL	NULL

products 5

14 20:58:57 SELECT * FROM products WHERE price > 1000 ... 4 row(s) returned

Q9. Show Electronics products with stock less than 20. Use `AND`.

```
51 •  SELECT *
52    FROM products
53    WHERE category = 'Electronics'
54    AND stock < 20;
55
```

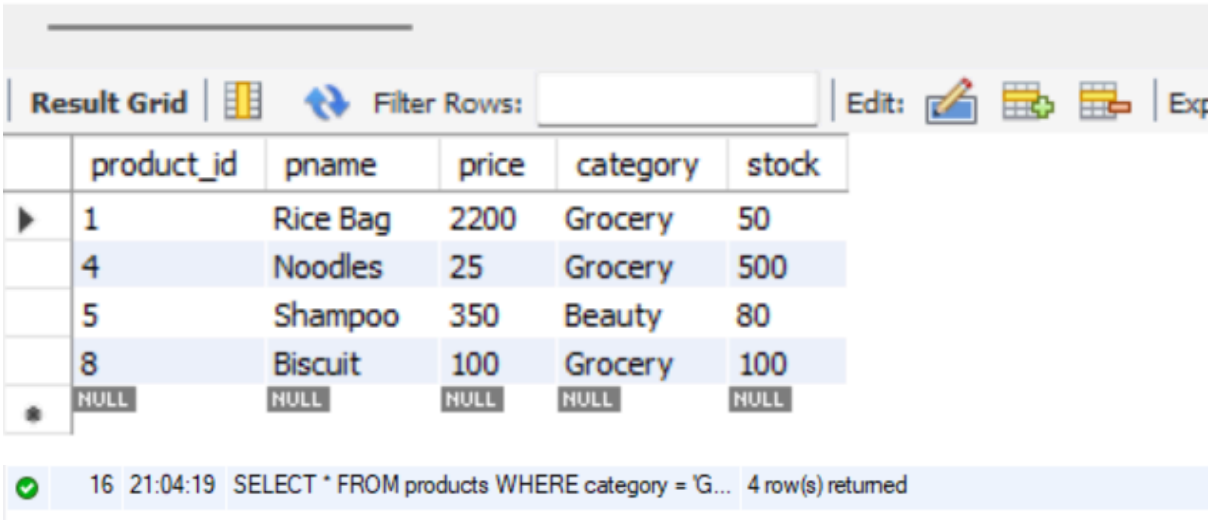
Result Grid | Filter Rows: | Edit: | Export/In

product_id	pname	price	category	stock
2	Laptop	85000	Electronics	10
6	TV	55000	Electronics	8
NULL	NULL	NULL	NULL	NULL

15 21:02:00 SELECT * FROM products WHERE category = 'E... 2 row(s) returned

Q10. Show products in Grocery OR products costing less than 500. Use `OR`.

```sql
SELECT *
FROM products
WHERE category = 'Grocery'
OR price < 500;
```



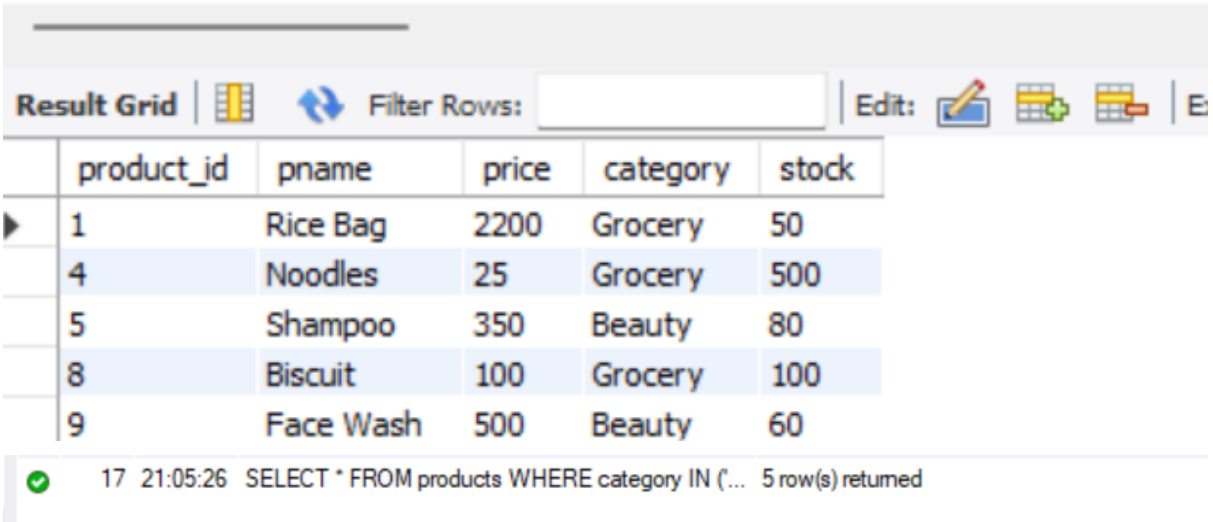
Result Grid | Filter Rows: | Edit:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
8	Biscuit	100	Grocery	100
NULL	NULL	NULL	NULL	NULL

16 21:04:19 SELECT * FROM products WHERE category = 'G... 4 row(s) returned

Q11. Show products in the Grocery and Beauty categories using `IN`.

```sql
SELECT *
FROM products
WHERE category IN ('Grocery', 'Beauty');
```



Result Grid | Filter Rows: | Edit:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
8	Biscuit	100	Grocery	100
9	Face Wash	500	Beauty	60

17 21:05:26 SELECT * FROM products WHERE category IN ('... 5 row(s) returned

Q12. Show products with price BETWEEN 300 and 40000. Is 350 included? Why?

```
66 •    SELECT *
67      FROM products
68      WHERE price BETWEEN 300 AND 40000;
```

Result Grid | Filter Rows: | Edit: | Export/

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
3	Mobile	30000	Electronics	25
5	Shampoo	350	Beauty	80
9	Face Wash	500	Beauty	60

18 21:06:52 SELECT * FROM products WHERE price BETW... 4 row(s) returned

Q13. Show products whose name starts with M.

```
71 •    SELECT *
72      FROM products
73      WHERE pname LIKE 'M%';
```

Result Grid | Filter Rows: | Edit: | Export/Im

product_id	pname	price	category	stock
3	Mobile	30000	Electronics	25

19 21:08:05 SELECT * FROM products WHERE pname LIKE '... 1 row(s) returned

Q14. Show products whose name contains the letter o.

```
76 •    SELECT *
77      FROM products
78      WHERE pname LIKE '%o%';
```

Result Grid | Filter Rows: | Edit: | E

product_id	pname	price	category	stock
2	Laptop	85000	Electronics	10
3	Mobile	30000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80

```
20  21:09:29  SELECT * FROM products WHERE pname LIKE '...  4 row(s) returned
```

Q15. Show the 2 most expensive products. Which two keywords work together?

```
80  SELECT *
81  FROM products
82  ORDER BY price DESC
83  LIMIT 2;
```

Result Grid | Filter Rows: | Edit:

product_id	pname	price	category	stock
2	Laptop	85000	Electronics	10
6	TV	55000	Electronics	8

```
21  21:12:25  SELECT * FROM products ORDER BY price DES...  2 row(s) returned
```

Q16. Change the Laptop price to 79000. Update using its ID.

```
85  UPDATE products
86  SET price = 79000
87  WHERE product_id = 2;
```

```
22  21:14:47  UPDATE products SET price = 79000 WHERE pr...  1 row(s) affected Rows matched: 1  Changed: 1  ...
```

Check

```
89  SELECT * FROM products
90  WHERE product_id = 2;
```

Result Grid | Filter Rows: | Edit:

product_id	pname	price	category	stock
2	Laptop	79000	Electronics	10

```
23  21:15:51  SELECT * FROM products WHERE product_id = ...  1 row(s) returned
```

Q17. Reduce the price of every Electronics product by 10%.

```sql
UPDATE products
SET price = price * 0.9
WHERE category = 'Electronics';

SET SQL_SAFE_UPDATES = 1;

SELECT * FROM products;
```

Result Grid | Filter Rows: | Edit:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	50
2	Laptop	71100	Electronics	10
3	Mobile	27000	Electronics	25
4	Noodles	25	Grocery	500
5	Shampoo	350	Beauty	80
6	TV	49500	Electronics	8

26	22:55:34	UPDATE products SET price = price * 0.9 WHER...	3 row(s) affected Rows matched: 3 Changed: 3	
27	22:55:41	SET SQL_SAFE_UPDATES = 1	0 row(s) affected	
28	22:55:48	SELECT * FROM products LIMIT 0, 1000	9 row(s) returned	

Q18. Ten Rice Bags were sold. Decrease the stock by 10 with one query.

```sql
SET SQL_SAFE_UPDATES = 0;

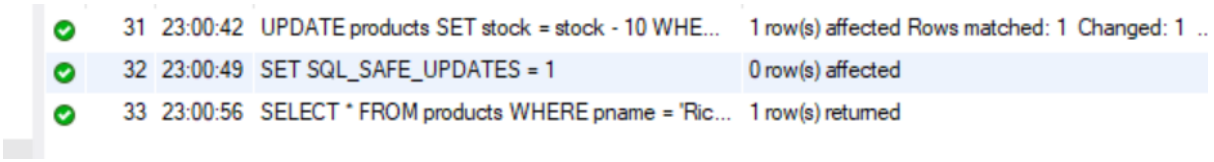
UPDATE products
SET stock = stock - 10
WHERE pname = 'Rice Bag';

SET SQL_SAFE_UPDATES = 1;

SELECT * FROM products
WHERE pname = 'Rice Bag';
```

Result Grid | Filter Rows: | Edit:

product_id	pname	price	category	stock
1	Rice Bag	2200	Grocery	40



✓	31	23:00:42	UPDATE products SET stock = stock - 10 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 ..
✓	32	23:00:49	SET SQL_SAFE_UPDATES = 1	0 row(s) affected
✓	33	23:00:56	SELECT * FROM products WHERE pname = 'Ric...	1 row(s) returned